



TIME SERIES ANALYSIS

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Outline

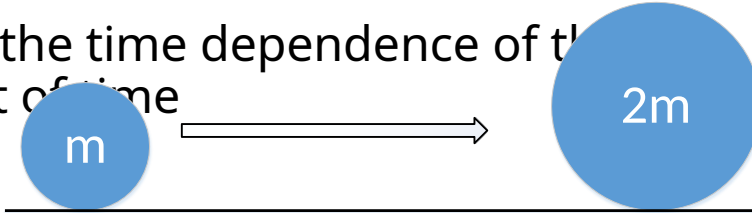
- Classification of Processes
- Why
Probability/Randomness?
- Randomness in real life
- Randomness in
realisations
- Ensembles

Classification of Processes

- Based on the nature of the processes, they can be classified as:

- Deterministic

- No uncertainty about the time dependence of the process
- Conservation laws



- Stochastic/Random

- Processes depend upon number of unknown factors, and hence, not completely predictable
- Most real life processes – throwing of die, detection of an encoded signal at receiver, weather forecasting, etc.

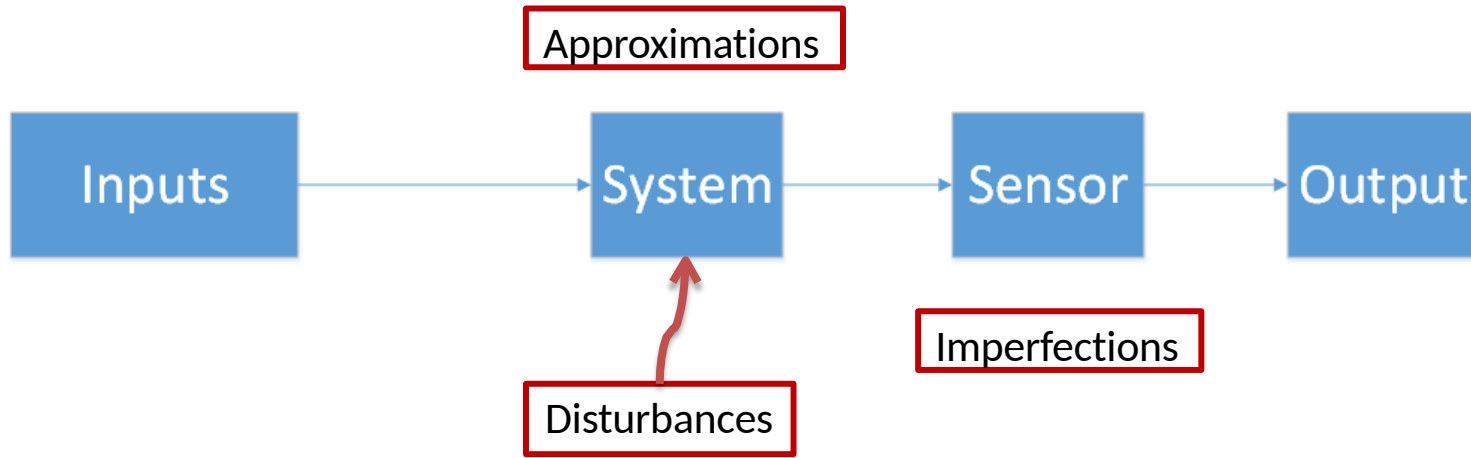
So how to analyze random processes??

— The concept of Probability





Why Probability/Randomness?



- No mathematical model is perfect
- Dynamic systems are driven by inputs as well as disturbances
- Sensors do not provide perfect data



Randomness – Examples

Scenario: Detecting a digitally modulated

- Input signal Received
: signal
- Sensor Demodulator

Randomness:

- Modeling response of demodulator System approximations
- Electromagnetic interference from weather, other channels, etc Disturbances
- Electronic noise in the demodulator Sensor imperfections



Randomness – Examples

Scenario: Measuring temperature of a room

- Inputs : Wind from fan, AC
- Sensor: Thermocouple

Randomness:

- Modeling the variation of temperature in relation to surroundings
- Movement of people into and out of the room (opening and closing of doors)
- Electronic noise in thermocouple

System approximations

Disturbances

Sensor imperfections

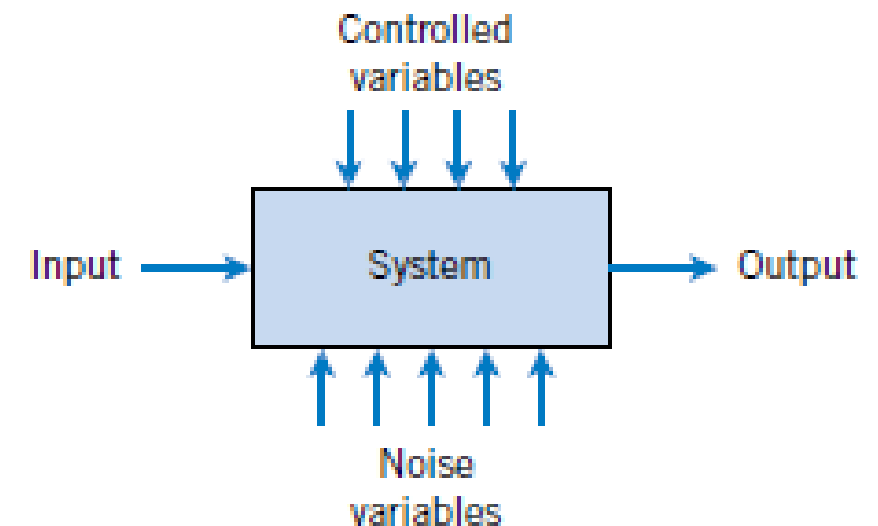
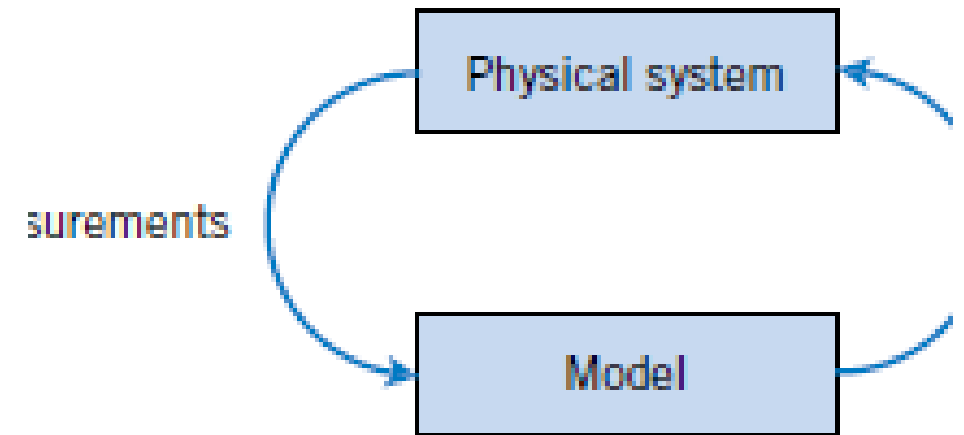


Random Processes

- Cannot be completely mathematically described
- Any real life experiment – a random process
- Outcome(s) of experiment – random variable(s)
- Measuring temperature – random process
- Temperature reading from sensor(s) – random variable(s)
- Thus, random processes can be viewed as data
processes generating (random
phenomena

Definitions

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**
- If we measure the current in a thin copper wire, we are conducting an experiment. However, in day-to-day repetitions of the measurement the results can differ slightly because of small variations in variables that are not controlled in our experiment, including changes in ambient temperatures, slight variations in gauge and small impurities in the chemical composition of the wire if different locations are selected, and current source drifts. Consequently, this experiment (as well as many we conduct) is said to have a **random component**.



Definitions Contd...

- The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as S .
 - Example: Consider an experiment in which you select a moulded plastic part, such as a connector, and measure its thickness. It might be convenient to define the sample space as simply the positive real line.
- A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes. A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.
- An **event (E)** is a subset of the sample space of a random experiment.

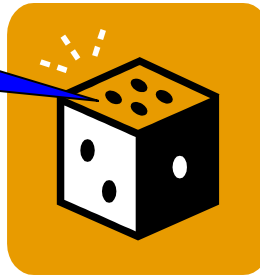
Events, Sample Spaces and Probability

- A **sample point** is the most basic outcome of an experiment.

A Head



A four



An Ace





Events, Sample Spaces and Probability

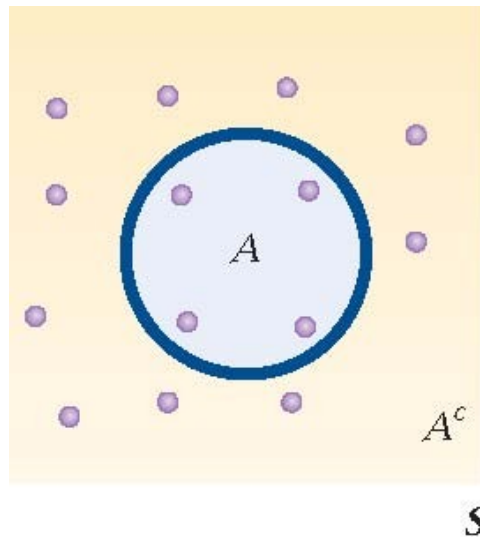
- A **sample space** of an experiment is the collection of all sample points.
 - Roll a single die:

$S: \{1, 2, 3, 4, 5, 6\}$



Events, Sample Spaces and Probability

- **Sample points and sample spaces** are often represented with Venn diagrams.





Events, Sample Spaces and Probability

- Probability Rules for Sample Points

- All probabilities must be between 0 and 1.

$$0 \leq p_i \leq 1$$

- The probabilities of *all* the sample points must sum to 1.

$$\sum_{i=1}^n p_i = 1$$



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Events, Sample Spaces and Probability

- Probability Rules for Sample Points

- All probabilities must be between 0 and 1

$$0 \leq p_i \leq 1$$

0 indicates a rare outcome and 1 a certain outcome.

- The probabilities of *all* the sample points must sum to 1

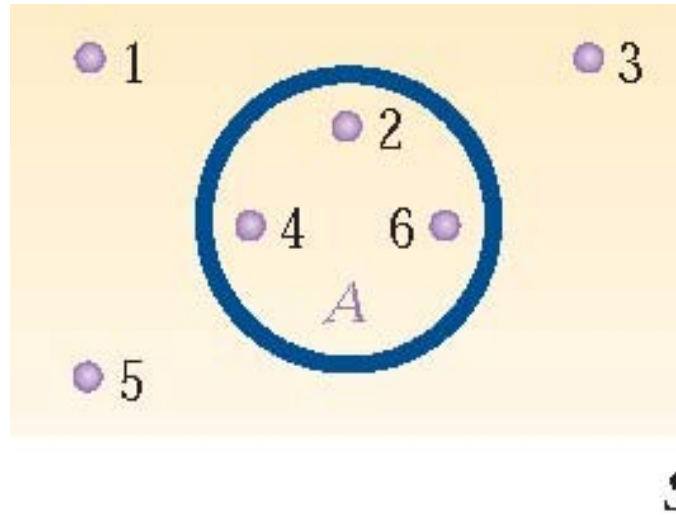
$$\sum_{i=1}^n p_i = 1$$

Something has to happen.

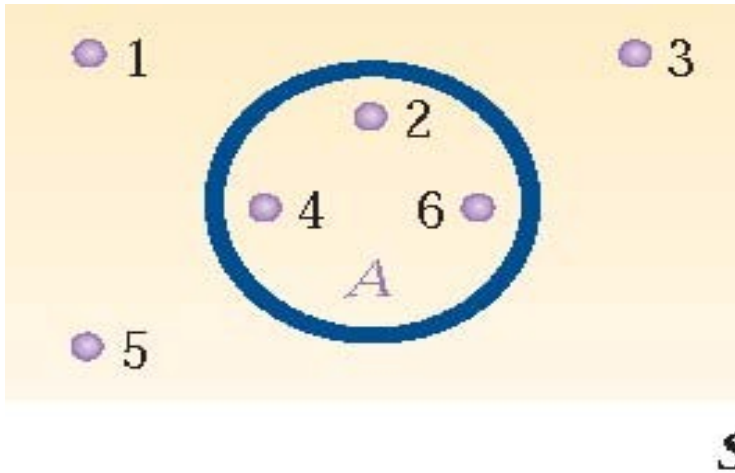


Events, Sample Spaces and Probability

- An **event** is a specific collection of sample points:
 - Event A: Observe an even number.



Events, Sample Spaces and Probability



- The **probability of an event** is the sum of the probabilities of the sample points in the sample space for the event.
 - Event A: Observe an even number.
 - $P(A) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$

Events, Sample Spaces and Probability

- Calculating Probabilities for Events
 - Define the experiment.
 - List the sample points.
 - Assign probabilities to sample points.
 - Collect all sample points in the event of interest.
 - The sum of the sample point probabilities is the event probability.

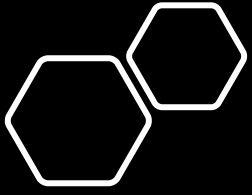


Events, Sample Spaces and Probability

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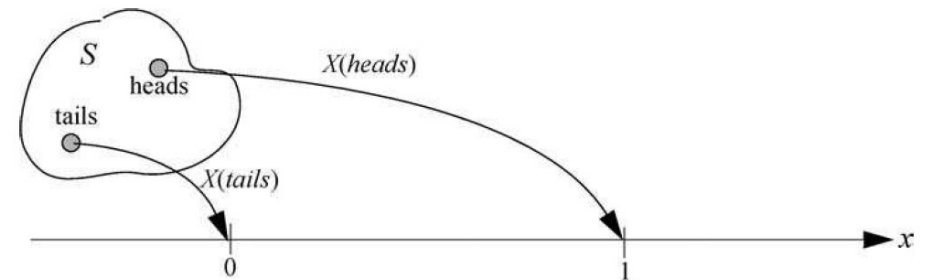
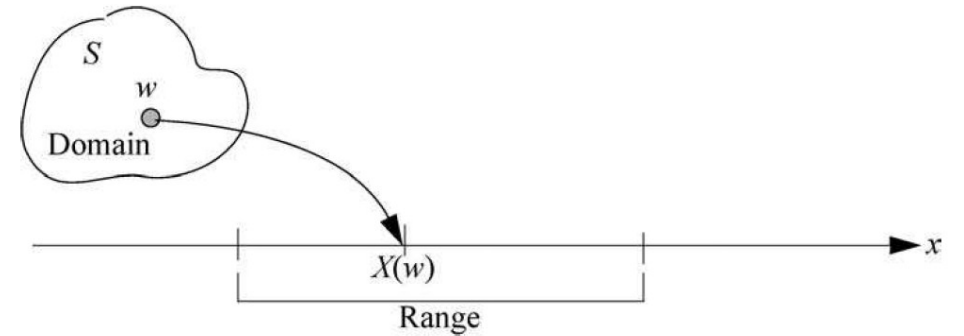
If the number of sample points gets too large, we need a way to keep track of how they can be combined for different events.

Probability Distribution



Random variables

- A **random variable** is a variable that assumes numerical values associated with the random outcome of an experiment, where one (and only one) numerical value is assigned to each sample point.





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Two Types of Random Variables

- **Discrete random variables**

- Number of sales
- Number of calls
- Shares of stock
- People in line
- Mistakes per page



- **Continuous random variables**

- Length
- Depth
- Volume
- Time
- Weight

Probability Distributions for Discrete Random Variables

- The **probability distribution** of a discrete random variable is a graph, table or formula that specifies the probability associated with each possible outcome the random variable can assume.
 - $p(x) \geq 0$ for all values of x
 - $\sum p(x) = 1$



x	$P(x)$
1	.30
2	.21
3	.15
4	.11
5	.07
6	.05
7	.04
8	.02
9	.02
10	.01

Probability Distributions for Discrete Random Variables

- Say a random variable x follows this pattern: $p(x) = (.3)(.7)^{x-1}$
- for $x > 0$.
 - This table gives the probabilities (rounded to two digits) for x between 1 and 10.

Discrete Random variables

- A voice communication system for a business contains 48 external lines. At a particular time, the system is observed, and some of the lines are being used. Let the random variable X denote the number of lines in use. Then, X can assume any of the integer values 0 through 48. When the system is observed, if 10 lines are in use, $x = 10$.
- Define the random variable X to be the number of contamination particles on a wafer in semiconductor manufacturing. Although wafers possess a number of characteristics, the random variable X summarizes the wafer only in terms of the number of particles

There is a chance that a bit transmitted through a digital transmission channel is received in error. Let X equal the number of bits in error in the next four bits transmitted. The possible values for X are $\{0, 1, 2, 3, 4\}$.

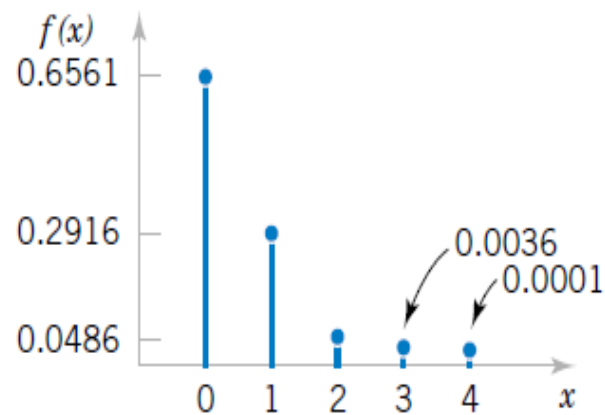


Figure 3-1 Probability distribution for bits in error.

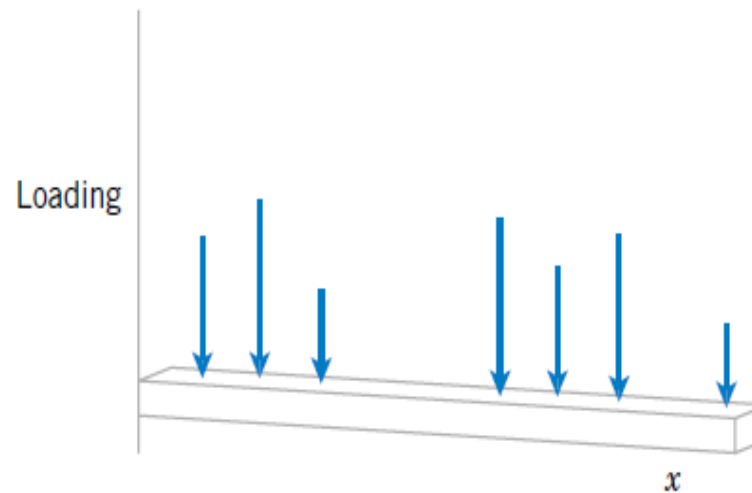


Figure 3-2 Loadings at discrete points on a long, thin beam.



For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, a **probability mass function** is a function such that

$$(1) \quad f(x_i) \geq 0$$

$$(2) \quad \sum_{i=1}^n f(x_i) = 1$$

$$(3) \quad f(x_i) = P(X = x_i) \quad (3-1)$$

Let the random variable X denote the number of semiconductor wafers that need to be analyzed in order to detect a large particle of contamination. Assume that the probability that a wafer contains a large particle is 0.01 and that the wafers are independent. Determine the probability distribution of X .

$$P(X = x) = \underbrace{P(aa \dots ap)}_{(x-1)a's} = 0.99^{x-1} (0.01), \quad \text{for } x = 1, 2, 3, \dots$$

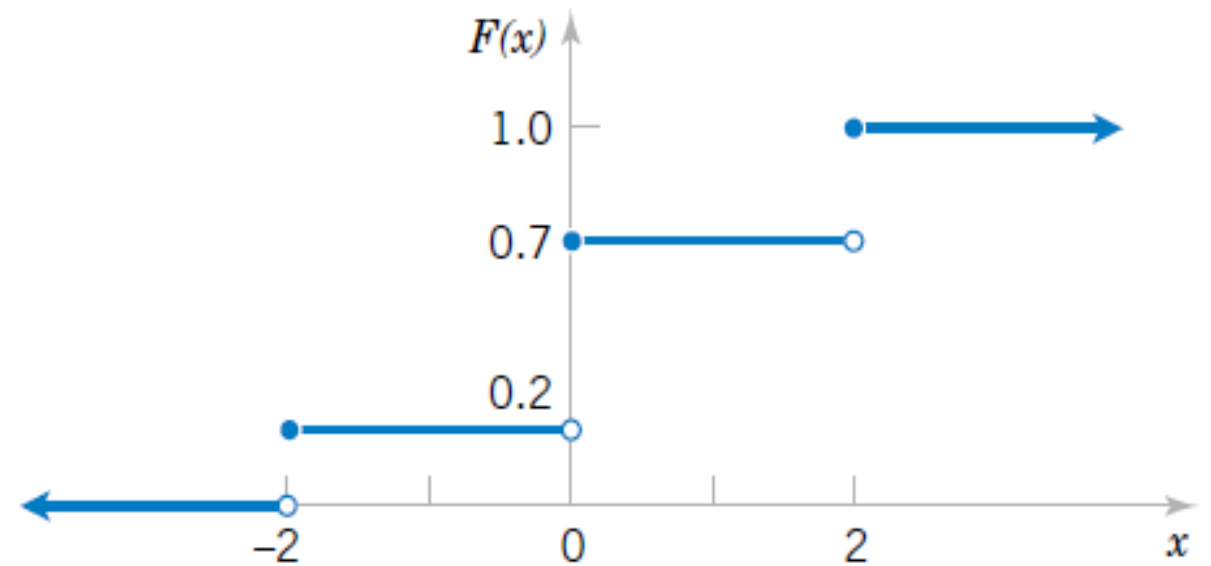
Cumulative distribution function

The **cumulative distribution function** of a discrete random variable X , denoted as $F(x)$, is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

For a discrete random variable X , $F(x)$ satisfies the following properties.

- (1) $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$
- (2) $0 \leq F(x) \leq 1$
- (3) If $x \leq y$, then $F(x) \leq F(y)$ (3-2)



$$f(-2) = 0.2 - 0 = 0.2$$

$$f(0) = 0.7 - 0.2 = 0.5$$

$$f(2) = 1.0 - 0.7 = 0.3$$



Expected Values of Discrete Random Variables

- The **mean**, or **expected value**, of a **discrete random variable** is

$$\mu = E(x) = \sum xp(x).$$





Expected Values of Discrete Random Variables

- The **variance** of a **discrete random variable** x is

$$\sigma^2 = E[(x - \mu)^2] = \sum (x - \mu)^2 p(x).$$

- The **standard deviation** of a **discrete random variable** x is

$$\sqrt{\sigma^2} = \sqrt{E[(x - \mu)^2]} = \sqrt{\sum (x - \mu)^2 p(x)}.$$



Expected Values of Discrete Random Variables

- In a roulette wheel in a U.S. casino, a \$1 bet on “even” wins \$1 if the ball falls on an even number (same for “odd,” or “red,” or “black”).
- The odds of winning this bet are 47.37%

$$P(\text{win } \$1) = .4737$$

$$P(\text{lose } \$1) = .5263$$

$$\mu = +\$1 \cdot .4737 - \$1 \cdot .5263 = - .0526$$

$$\sigma = .9986$$

On average, bettors lose about a nickel for each dollar they put down on a bet like this.

(These are the *best* bets for patrons.)



Population and Sample mean

Definition

A **population** consists of the totality of the observations with which we are concerned.

Definition

A **sample** is a subset of observations selected from a population.

Definition

A **statistic** is any function of the observations in a random sample.



Definition

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

$$E(\hat{\Theta}) = \theta \quad (7-1)$$

If the estimator is not unbiased, then the difference

$$E(\hat{\Theta}) - \theta \quad (7-2)$$

is called the **bias** of the estimator $\hat{\Theta}$.

Now, Sample mean is unbiased. Proof?

Variance of sample mean?



Sample variance

2. The following equality is helpful in many derivations,

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - a)^2 - n\{\bar{X} - a\}^2, \forall a.$$

This is because

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n \{(X_i - a) - (\bar{X} - a)\}^2 = \sum_{i=1}^n [(X_i - a)^2 - 2(X_i - a)(\bar{X} - a) + (\bar{X} - a)^2].$$

Note that $\sum_{i=1}^n (X_i - a)(\bar{X} - a) = n(\bar{X} - a)(\bar{X} - a)$, the result in equation (3) follows.

3. There are two equivalent ways of expressing variance, they are

$$Var(X_1) = E\{X - E(X)\}^2 = E(X^2) - \{E(X)\}^2.$$

Now,

$$E(\bar{X}) = n^{-1} \sum_{i=1}^n E(X_i) = n^{-1} \cdot n\mu = \mu.$$



$$E(s^2) = E\left\{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}\right\} = \frac{1}{n-1} E\left\{\sum_{i=1}^n (X_i - \bar{X})^2\right\} = \frac{\sum_{i=1}^n E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2}{n-1}.$$

Note that $E(X_i - \mu)^2 = \text{Var}(X_i) = \sigma^2$ and $E(\bar{X} - \mu)^2 = \text{Var}(\bar{X}) = n^{-1}\text{Var}(X_i) = n^{-1}\sigma^2$, we then have

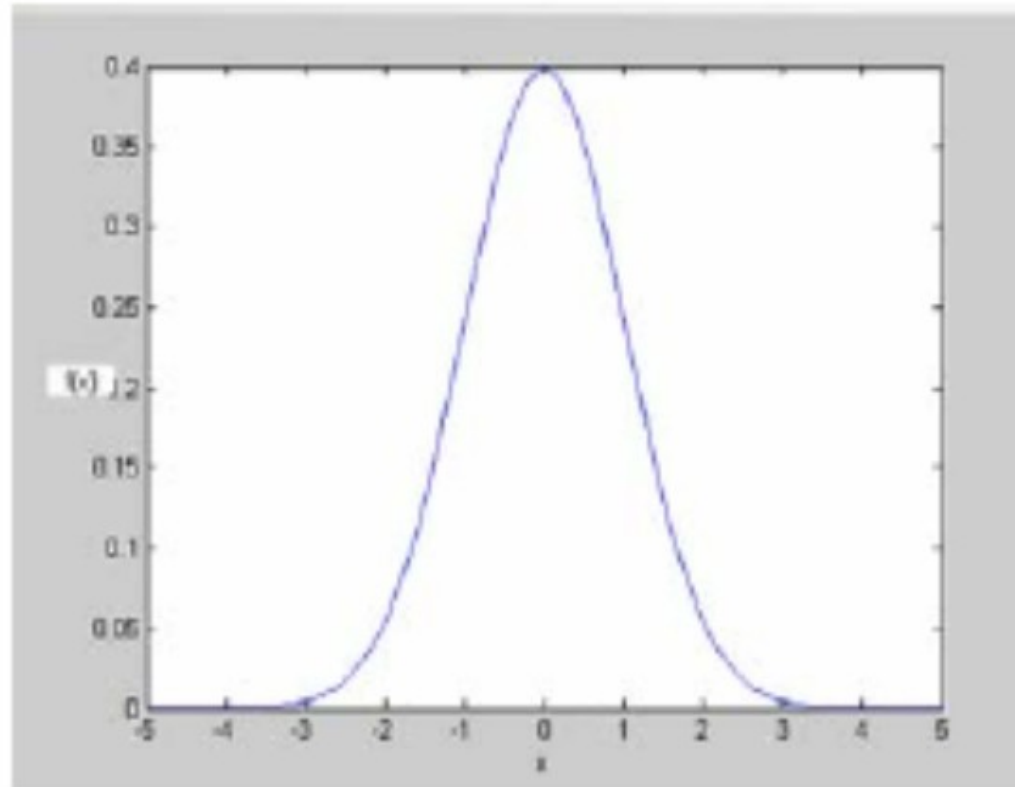
$$E(s^2) = \frac{n\sigma^2 - n \cdot n^{-1}\sigma^2}{n-1} = \sigma^2.$$



Normal Distribution

- The pdf of a normally distributed RV is given by:

$$f(x) = \frac{1}{\sigma_x \sqrt{2\pi}} e^{\left(-\frac{1}{2} \frac{(x-\mu_x)^2}{\sigma_x^2}\right)}$$



Standard normal distribution: zero mean and unit variance.



The Normal Distribution

- Closely approximates many situations
 - Perfectly symmetrical around its mean
- The probability density function $f(x)$:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{[(x-\mu)/\sigma]^2}{2}}$$

μ = the mean of x

σ = the standard deviation of x

π = 3.1416...

e = 2.71828 ...



The Normal Distribution

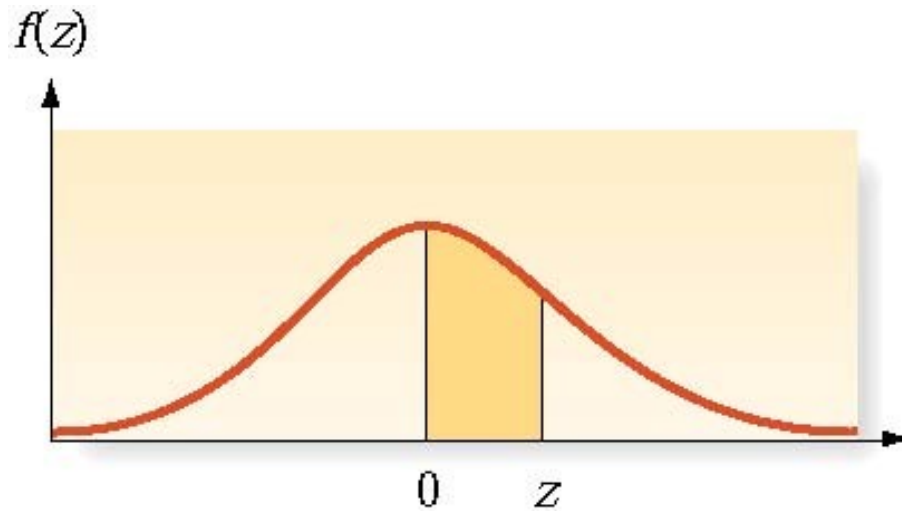
- Each combination of μ and σ produces a unique normal curve
- The **standard normal curve** is used in practice, based on the standard normal random variable z ($\mu = 0, \sigma = 1$), with the probability distribution

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

The probabilities for z are given in Table IV



The Normal Distribution



$$P(0 < z < 1.00) = .3413$$

$$P(-1.00 < z < 0) = .3413$$

$$\begin{aligned} P(-1 < z < 1) &= .3413 + .3413 \\ &= .6826 \end{aligned}$$

$$\begin{aligned} P(1 < z < 1.25) &= \\ P(0 < z < 1.25) - P(0 < z < 1.00) \\ &= .3944 - .3413 = .0531 \end{aligned}$$



The Normal Distribution

For a normally distributed random variable x , if we know μ and σ ,

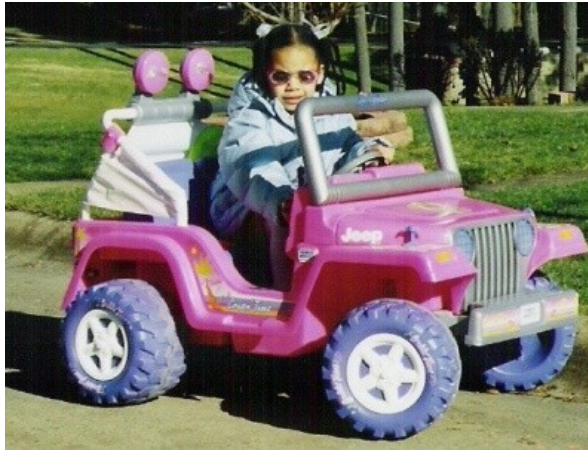
$$z_i = \frac{x_i - \mu}{\sigma}$$

So *any* normally distributed variable can be analyzed with this single distribution



The Normal Distribution

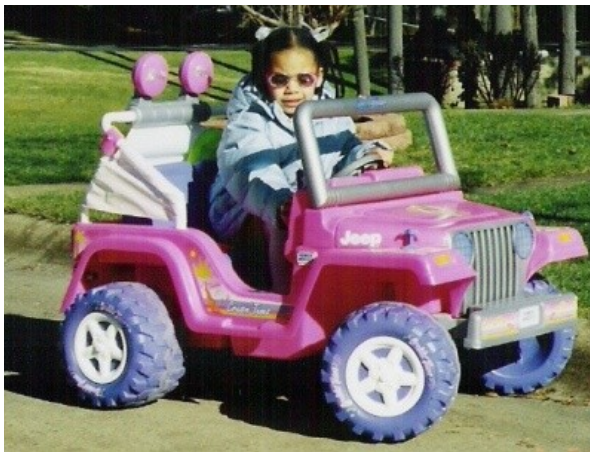
- Say a toy car goes an average of 3,000 yards between recharges, with a standard deviation of 50 yards (i.e., $\mu = 3,000$ and $\sigma = 50$)
- What is the probability that the car will go more than 3,100 yards without recharging?





The Normal Distribution

- Say a toy car goes an average of 3,000 yards between recharges, with a standard deviation of 50 yards (i.e., $\mu = 3,000$ and $\sigma = 50$)
- What is the probability that the car will go more than 3,100 yards without recharging?



$$\begin{aligned} P(x > 3100) &= P\left(z > \frac{3100 - 3000}{50}\right) = \\ P(z > 2.00) &= 1 - P(z < 2.00) = \\ &= 1 - .5 - P(0 < z < 2.00) = \\ &= 1 - .5 - .4772 = .0228 \end{aligned}$$



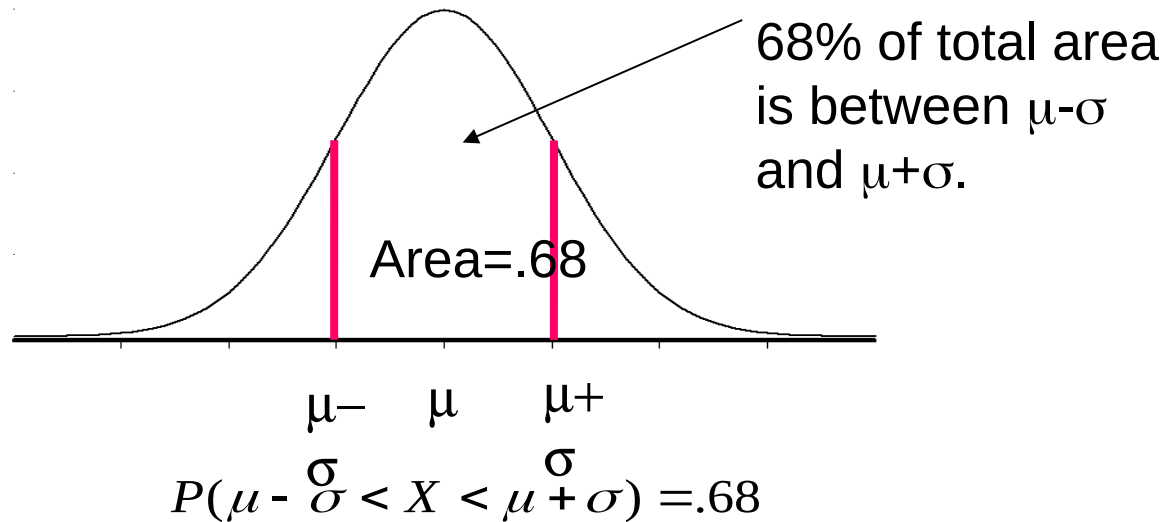
The Normal Distribution

- To find the probability for a normal random variable ...
 - Sketch the normal distribution
 - Indicate x 's mean
 - Convert the x variables into z values
 - Put both sets of values on the sketch, z below x
 - Use Table IV to find the desired probabilities



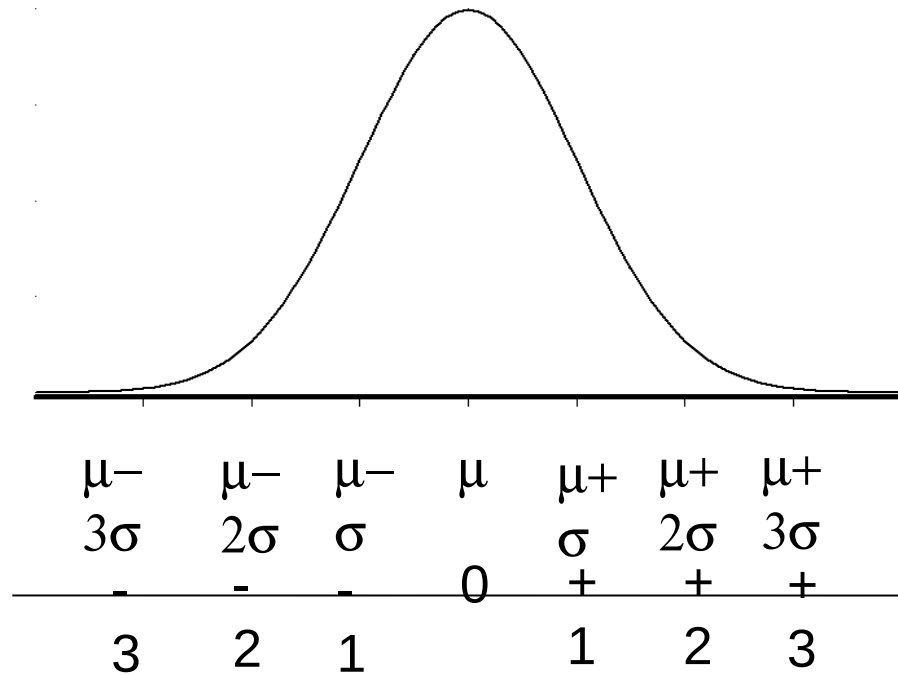
Normal Distribution

A symmetric distribution defined on the range $-\infty$ to $+\infty$ whose shape is defined by two parameters, the **mean**, denoted μ , that centers the distribution, and the **standard deviation**, σ , that determines the spread of the distribution.





Standard Normal Distribution

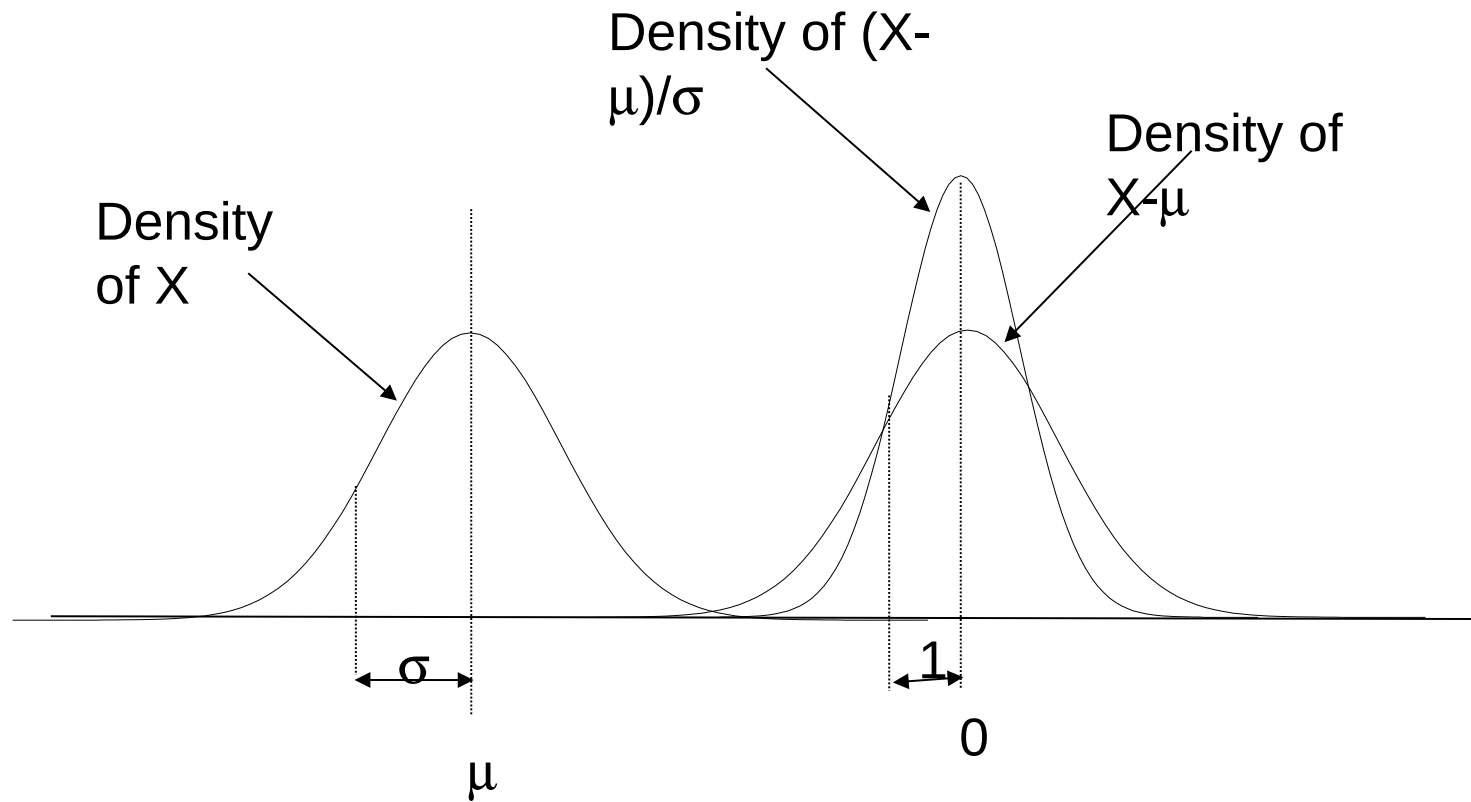


All normal random variables can be related back to the **standard normal random variable**.

A Standard Normal random variable has mean 0 and standard deviation 1.



Illustration





Notation

Suppose X has a normal distribution with mean μ and standard deviation σ , denoted $X \sim N(\mu, \sigma)$.

Then a new random variable defined as $Z = (X - \mu) / \sigma$, has the standard normal distribution, denoted $Z \sim N(0, 1)$.

$$Z = (X - \mu) / \sigma$$

To **standardize** we subtract the mean and divide by the standard deviation.

$$\sigma Z + \mu = X$$

To create a random variable with specific mean and standard deviation, we start with a standard normal deviate, multiply it by the target standard deviation, and then add the target mean.

Why is this important? Because in this way, the probability of any event on a normal random variable with any given mean and standard deviation can be computed from tables of the standard normal distribution.



Relating Any Normal RV to a Standard Normal RV

$$X \sim N(\mu, \sigma)$$

$$Z \sim N(0,1)$$

$$X = \sigma Z + \mu$$

$$P(X < x_0) = P(\sigma Z + \mu < x_0)$$

$$P(\sigma Z < x_0 - \mu)$$

$$P\left(Z < \frac{x_0 - \mu}{\sigma}\right)$$

This probability can be found in a table of standard normal probabilities (Table 1 in Ott and Longnecker)

This value is just a number, usually between ± 4

Other Useful Relationships

$$P(Z < z_0) = 1 - P(Z > z_0)$$

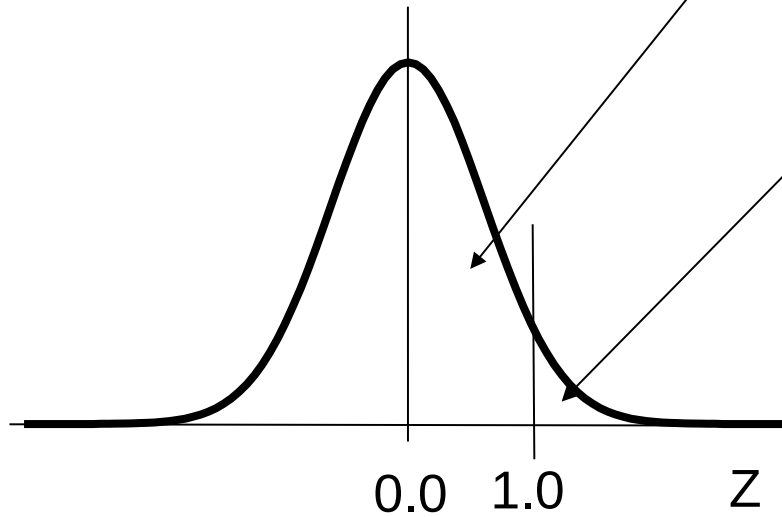
Probability of complementary events.

$$P(Z < -z_0) = P(Z > +z_0)$$

Symmetry of the normal distribution.



Normal Table



Ott & Longnecker, Table 1 page 676, gives areas left of z . This table from a previous edition gives areas right of z .

TABLE 2 Upper-tail Areas for the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.00	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.10	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.20	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.30	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.40	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.50	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.60	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.70	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.80	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.90	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.00	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.10	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.20	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.30	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.40	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.50	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.60	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.70	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.80	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.90	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.00	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.10	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.20	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.30	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.40	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.50	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.60	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.70	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.80	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.90	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.00	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010

z	Area
3.500	.00023263
4.000	.00003167
4.500	.00000340
5.000	.00000029

Source: Computed by J. W. Stegeman using SAS.

Page A-3 OTT 1998
An Introduction To
STATISTICAL Methods
and Data Analysis

Find $P(2 < X < 4)$ when $X \sim N(5, 2)$.
The standardization equation for X is:
 $Z = (X - \mu) / \sigma = (X - 5) / 2$

when $X=2$, $Z = -3/2 = -1.5$

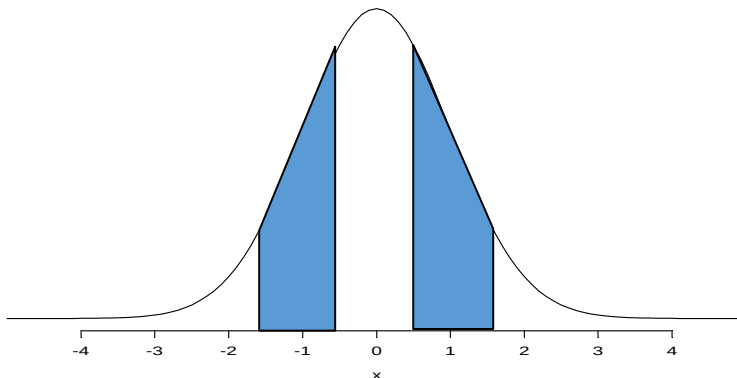
when $X=4$, $Z = -1/2 = -0.5$

$$P(2 < X < 4) = P(X < 4) - P(X < 2)$$

$$P(X < 2) = P(Z < -1.5) \\ = P(Z > 1.5) \text{ (by symmetry)}$$

$$P(X < 4) = P(Z < -0.5) \\ = P(Z > 0.5) \text{ (by symmetry)}$$

$$P(2 < x < 4) = P(X < 4) - P(X < 2) \\ = P(Z > 0.5) - P(Z > 1.5) \\ = 0.3085 - 0.0668 = 0.2417$$



Using a Normal Table

TABLE 2 Upper-tail Areas for the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.00	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.10	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.20	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.30	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.40	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.50	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.60	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.70	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.80	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.90	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.00	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.10	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.20	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.30	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.40	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.50	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.60	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.70	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.80	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.90	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.00	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.10	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.20	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.30	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.40	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.50	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.60	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.70	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.80	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.90	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.00	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010

z	Area
3.500	.00023263
4.000	.00003167
4.500	.00000340
5.000	.00000029

Source: Computed by J. W. Stegeman using SAS.

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An Introduction To
STATISTICAL Methods
and Data Analysis



Properties of the Normal Distribution

- Symmetric, bell-shaped density function.
- 68% of area under the curve between $\mu \pm \sigma$.
- 95% of area under the curve between $\mu \pm 2\sigma$.
- 99.7% of area under the curve between $\mu \pm$

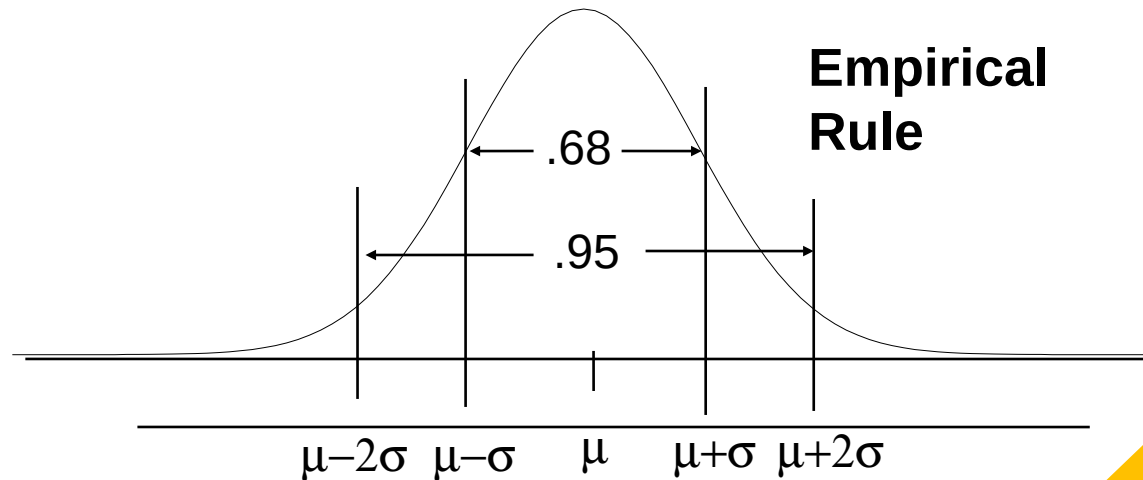
3σ .

$$Y \sim N(\mu, \sigma)$$

$$Y - \mu \sim N(0, \sigma)$$

$$\frac{Y - \mu}{\sigma} \sim N(0, 1)$$

Standard Normal Form

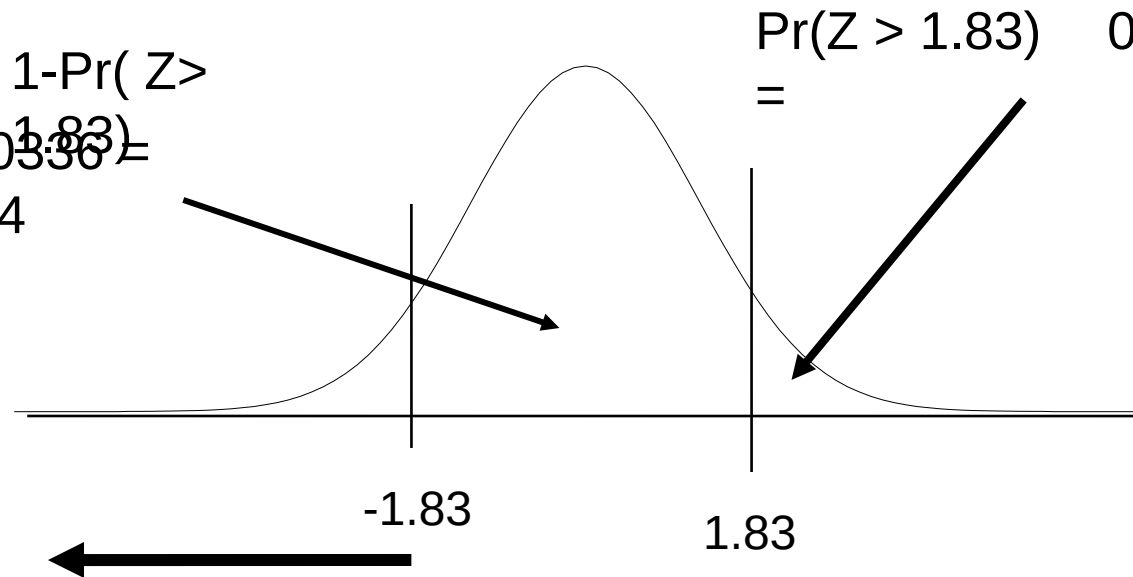




Probability Problems

Using symmetry and the fact that the area under the density curve is 1.

$$\Pr(Z < 1.83) = 1 - \Pr(Z > 1.83) = 1 - 0.0336 = 0.9664$$



$$\Pr(Z < -1.83) = \Pr(Z > 1.83) = 0.0336$$

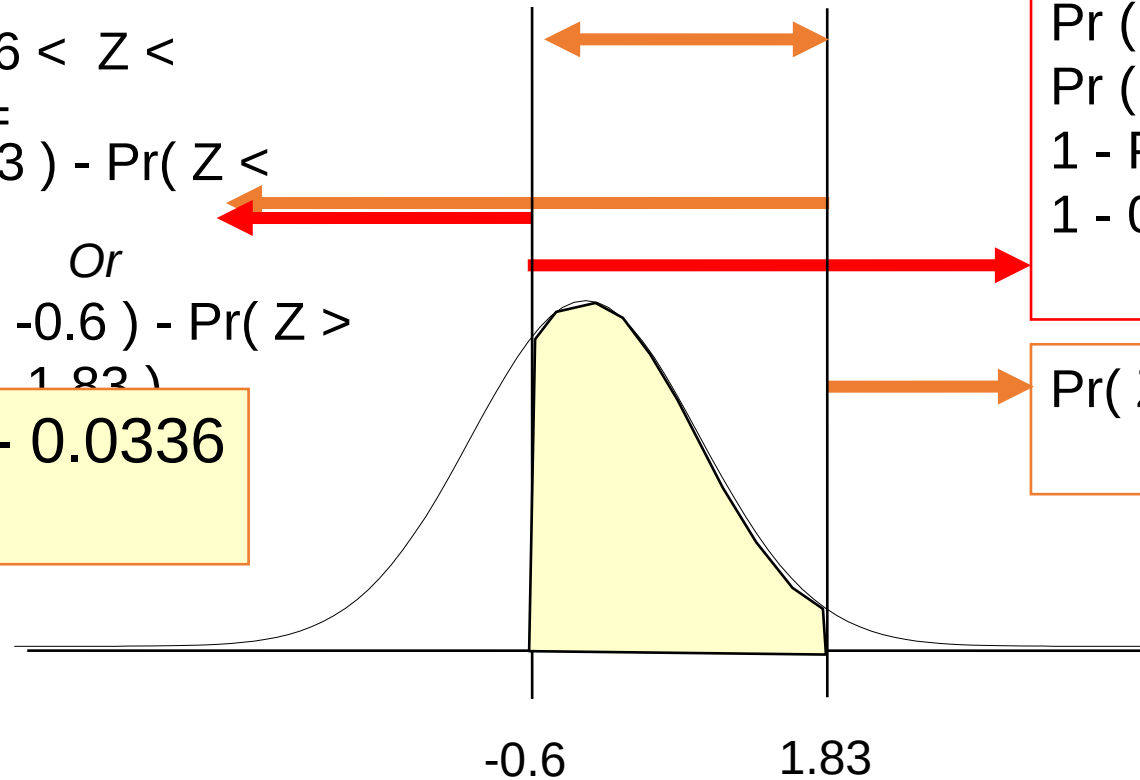
By
Symmetry

Probability Problems

Cutting out the tails.

$$\begin{aligned} & \Pr(-0.6 < Z < 1.83) = \\ & \Pr(Z < 1.83) - \Pr(Z < -0.6) \\ & \text{Or} \\ & \Pr(Z > -0.6) - \Pr(Z > 1.83) \end{aligned}$$

$$\begin{aligned} & = 0.7257 - 0.0336 \\ & = 0.6921 \end{aligned}$$



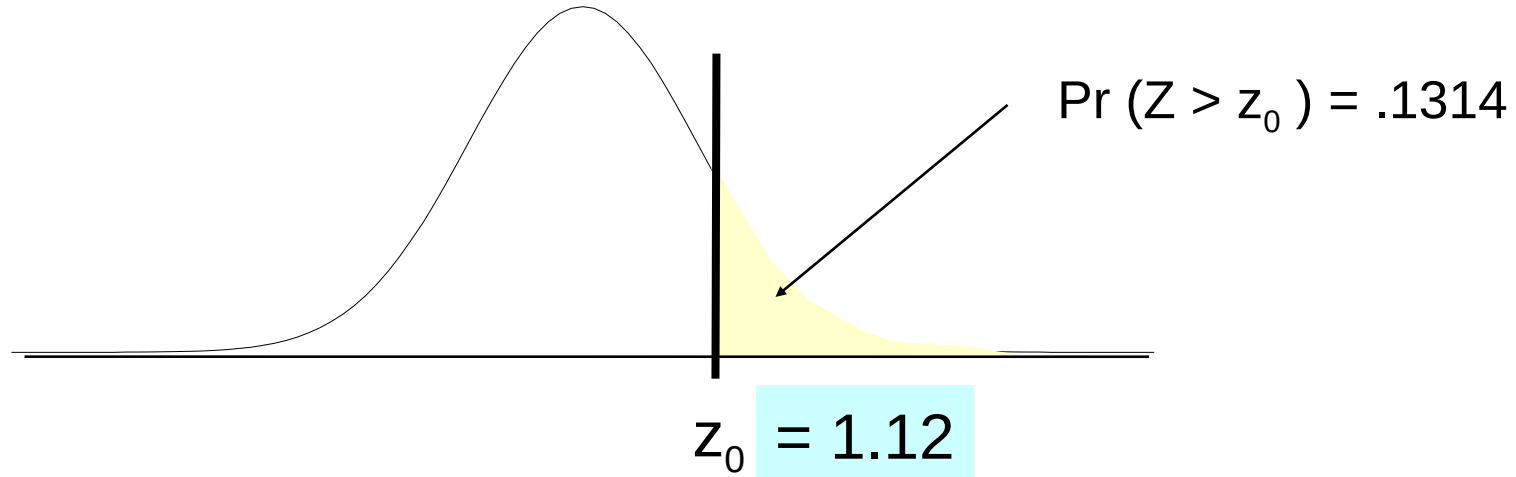
$$\begin{aligned} \Pr(Z > -0.6) &= \\ \Pr(Z < +0.6) &= \\ 1 - \Pr(Z > 0.6) &= \\ 1 - 0.2743 &= \\ 0.7257 \end{aligned}$$

$$\begin{aligned} \Pr(Z > 1.83) &= \\ 0.0336 \end{aligned}$$

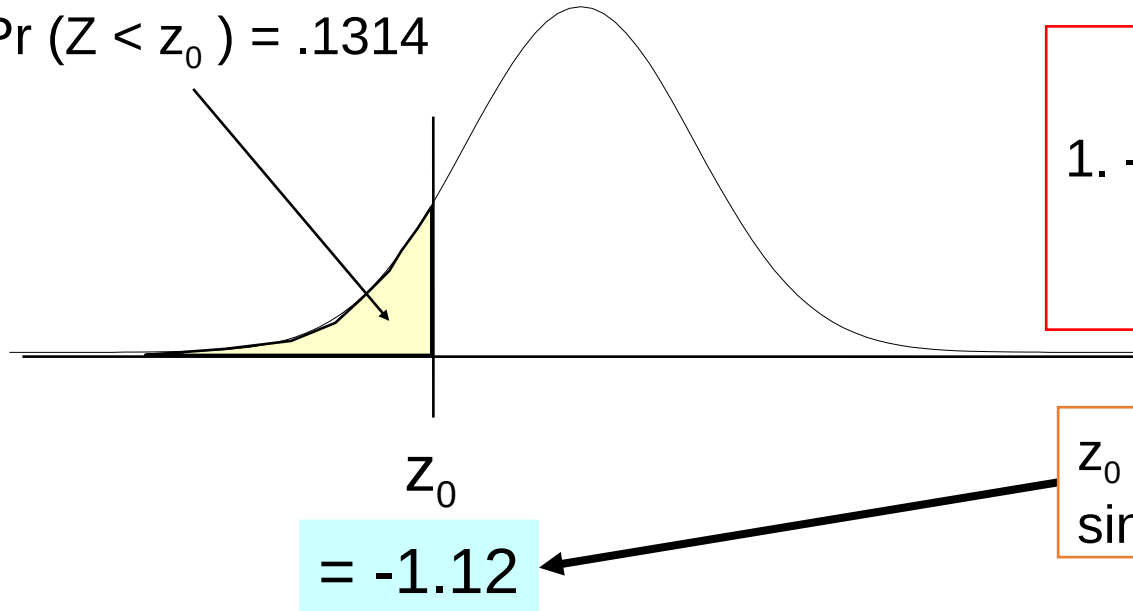


Given the Probability - What is Z_0 ?

Working backwards.



$\Pr(Z < z_0) = .1314$



$$\begin{aligned}\Pr(Z < z_0) &= 0.1314 \\ 1. - \Pr(Z > z_0) &= 0.1314 \\ \Pr(Z > z_0) &= 1 - 0.1314 \\ &= 0.8686\end{aligned}$$

z_0 must be negative,
since $\Pr(Z > 0.0) = 0.5$



Converting to Standard Normal Form

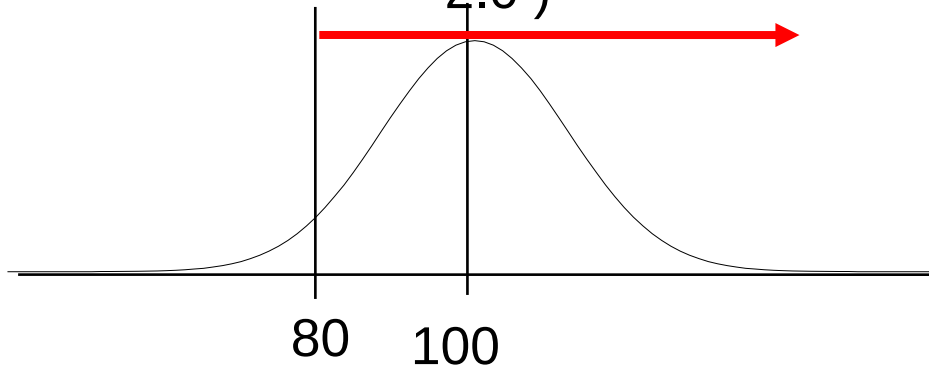
$$Z = \frac{X - \mu}{\sigma}$$

Suppose we have a random variable (say weight), denoted by W , that has a normal distribution with mean 100 and standard deviation 10.

$$W \sim N(100, 10)$$

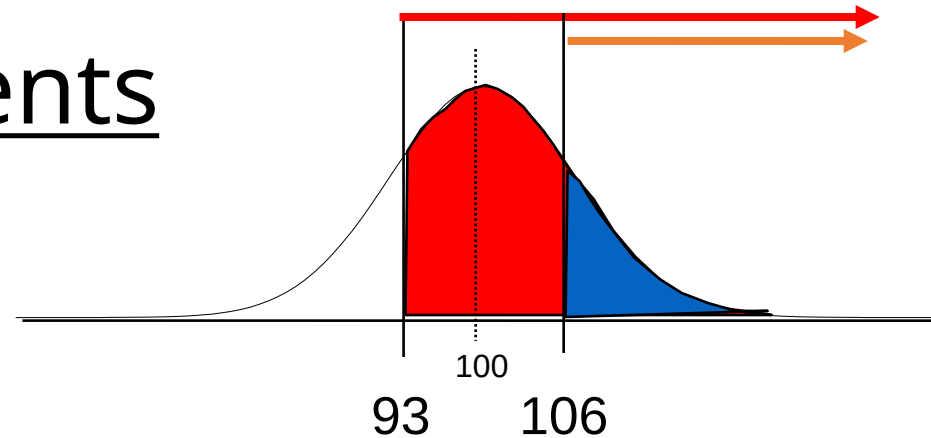
$$\begin{aligned} \Pr(W < 90) &= \Pr(Z < (90 - 100) / 10) = \Pr(Z < -1.0) \\ &= \Pr(Z > 1.0) = 0.1587 \end{aligned}$$

$$\begin{aligned} \Pr(W > 80) &= \Pr(Z > (80 - 100) / 10) = \Pr(Z > -2.0) \\ &= 1.0 - \Pr(Z < -2.0) \\ &= 1.0 - \Pr(Z > 2.0) \\ &= 1.0 - 0.0228 = 0.9772 \end{aligned}$$



Decomposing Events

$$W \sim N(100, 10)$$



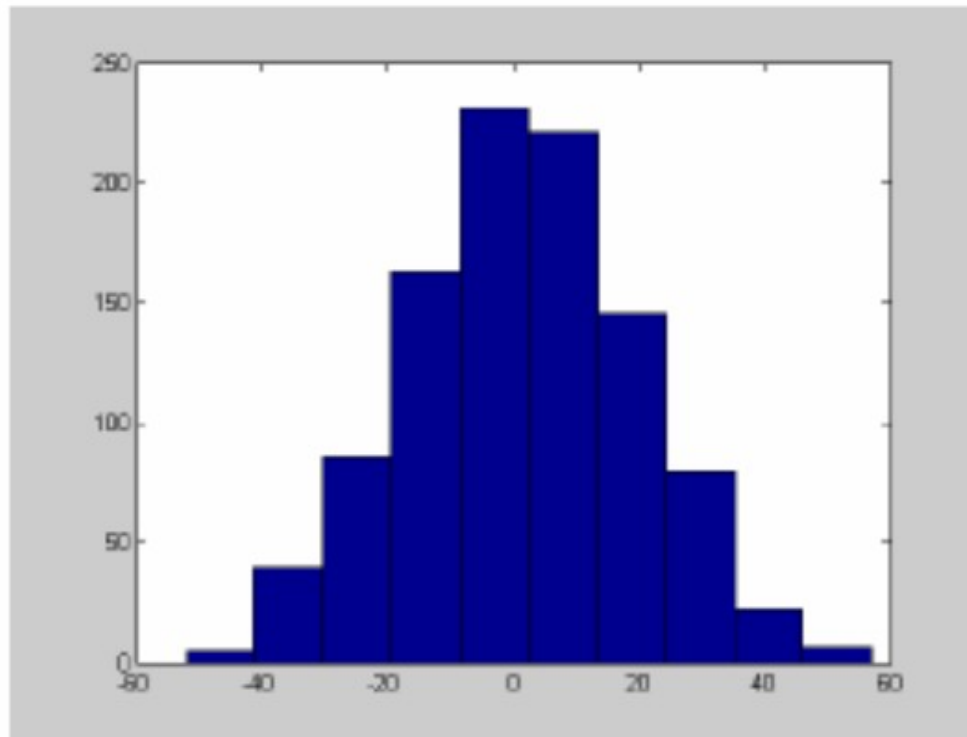
$$\begin{aligned} \Pr(93 < W < 106) &= \Pr(W > 93) - \Pr(W > 106) \\ &= \Pr[Z > (93 - 100) / 10] - \Pr[Z > (106 - 100) / 10] \\ &= \Pr[Z > -0.7] - \Pr[Z > +0.6] \\ &= 1.0 - \Pr[Z > +0.7] - \Pr[Z > +0.6] \\ &= 1.0 - 0.2420 - .2743 = 0.4837 \end{aligned}$$



Central Limit Theorem

Theorem

- The sum of several independent random variables tends to have a normal distribution regardless of the underlying distributions of the individual random variables.



Sum of 10 uniform
random numbers in
Range $[-10, 10]$
Histogram of 1000 samples

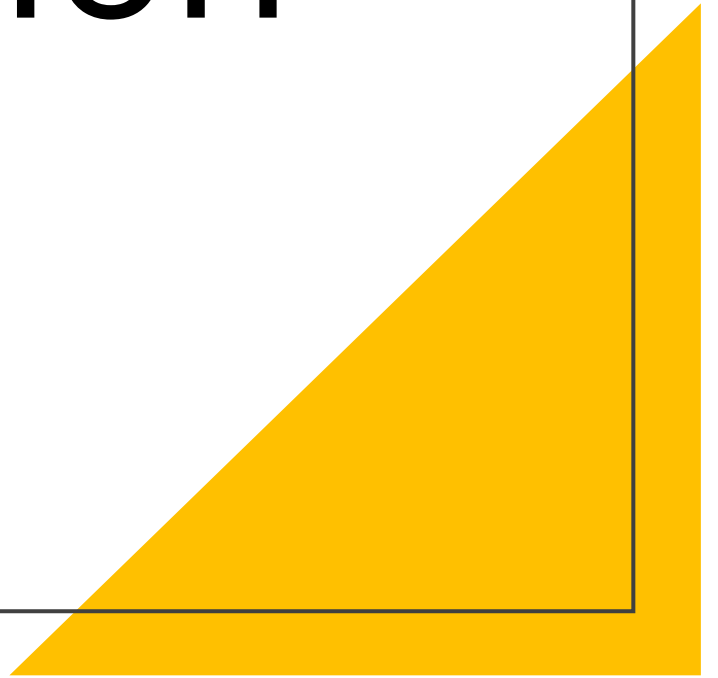
```
x1 = unifrnd(-10,10,1000,10);  
y = sum(x1,2);  
hist(y)
```



Aside: Caveat about CLT

- Central limit theorem works if original distribution are not heavy tailed
 - Need to have enough samples. Eg: with multipaths, if there is not rich enough scattering, the convergence to normal may have not happened yet
- Moments converge to limits
- Trouble with aggregates of “heavy tailed” distribution samples
- Rate of convergence to normal also varies with distributional skew, and dependence in samples
- Non-classical version of CLT for some cases (heavy tailed)...
 - Sum converges to stable Levy-noise (heavy tailed and long-range dependent auto-correlations)

3. Joint Distribution





Bivariate random variables

Definition: Bivariate Random Variable.

Given a random experiment with a sample space Ω , and a probability set function $P(\cdot)$ defined on its subsets; let there be a function X , defined on Ω , which assigns to each element $\omega \in \Omega$, one and only one ordered number pair $(X_1(\omega), X_2(\omega))$. This function, X , is called a two-dimensional, or bivariate *random variable*.

As with the single random variable case, associated with this two-dimensional random variable is a space, V , and a probability set function P_X induced by $X = (X_1, X_2)$, where V is defined as:

$$V = \{(x_1, x_2) : X_1(\omega) = x_1, X_2(\omega) = x_2; \omega \in \Omega\} \quad (5.1)$$

The most important point to note at this point is that the random variable space V involves X_1 and X_2 simultaneously; it is not merely a union of separate spaces V_1 for X_1 and V_2 for X_2 .



Consider two random variables X and Y defined on the same sample space. For example, X can denote the grade of a student and Y can denote the height of the same student. The joint cumulative distribution function (joint CDF) of X and Y

is given by

$$F_{XY}(x, y) = P[X \leq x, Y \leq y]$$

The pair (X, Y) is referred to as a *bivariate* random variable. If we define $F_X(x) = P[X \leq x]$ as the *marginal* CDF of X and $F_Y(y) = P[Y \leq y]$ as the *marginal* CDF of Y , then we define the random variables X and Y to be independent if

$$F_{XY}(x, y) = F_X(x)F_Y(y)$$

for every value of x and y .



Consider two random variables X and Y defined on the same sample space. For example, X can denote the grade of a student and Y can denote the height of the same student. The joint cumulative distribution function (joint CDF) of X and Y

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$$F_{XY}(x, y) = F_X(x)F_Y(y)$$

for every value of x and y .



5.2.1 Properties of the Joint CDF

As a probability function $F_{XY}(x, y)$ has certain properties, which include the following:

1. Since $F_{XY}(x, y)$ is a probability, $0 \leq F_{XY}(x, y) \leq 1$ for $-\infty < x < \infty$, $-\infty < y < \infty$.
2. If $x_1 \leq x_2$ and $y_1 \leq y_2$, then $F_{XY}(x_1, y_1) \leq F_{XY}(x_2, y_1) \leq F_{XY}(x_2, y_2)$. Similarly, $F_{XY}(x_1, y_1) \leq F_{XY}(x_1, y_2) \leq F_{XY}(x_2, y_2)$. This follows from the fact that $F_{XY}(x, y)$ is a nondecreasing function of x and y .
3. $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} F_{XY}(x, y) = F_{XY}(\infty, \infty) = 1$
4. $\lim_{x \rightarrow -\infty} F_{XY}(x, y) = F_{XY}(-\infty, y) = 0$
5. $\lim_{y \rightarrow -\infty} F_{XY}(x, y) = F_{XY}(x, -\infty) = 0$
6. $\lim_{x \rightarrow a^+} F_{XY}(x, y) = F_{XY}(a, y)$
7. $\lim_{y \rightarrow b^+} F_{XY}(x, y) = F_{XY}(x, b)$
8. $P[x_1 < X \leq x_2, Y \leq y] = F_{XY}(x_2, y) - F_{XY}(x_1, y)$
9. $P[X \leq x, y_1 < Y \leq y_2] = F_{XY}(x, y_2) - F_{XY}(x, y_1)$
10. If $x_1 \leq x_2$ and $y_1 \leq y_2$, then

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1) \geq 0$$

Also, the marginal CDFs are obtained as follows:

$$F_X(x) = F_{XY}(x, \infty)$$

$$F_Y(y) = F_{XY}(\infty, y)$$

Definition

The **joint probability mass function** of the discrete random variables X and Y , denoted as $f_{XY}(x, y)$, satisfies

- (1) $f_{XY}(x, y) \geq 0$
 - (2) $\sum_x \sum_y f_{XY}(x, y) = 1$
 - (3) $f_{XY}(x, y) = P(X = x, Y = y)$
- (5-1)

EXAMPLE 5-1

In the development of a new receiver for the transmission of digital information, each received bit is rated as *acceptable*, *suspect*, or *unacceptable*, depending on the quality of the received signal, with probabilities 0.9, 0.08, and 0.02, respectively. Assume that the ratings of each bit are independent.

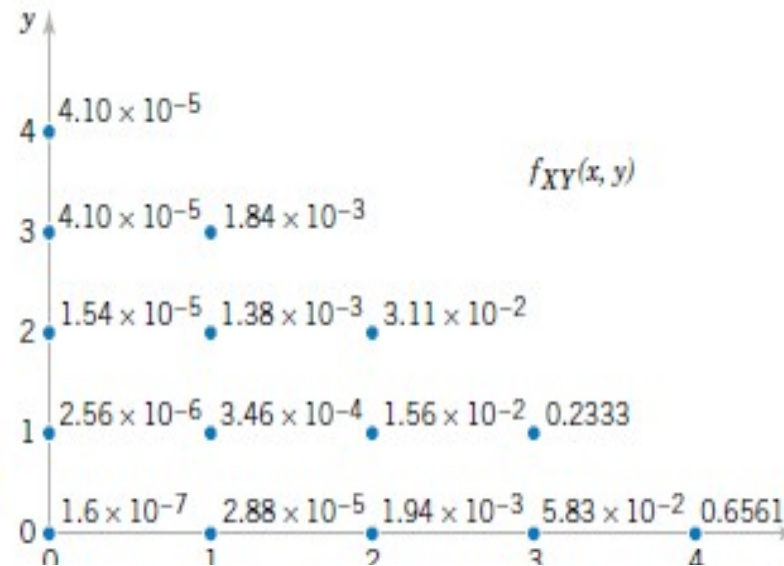
In the first four bits transmitted, let

X denote the number of acceptable bits

Y denote the number of suspect bits

Then, the distribution of X is binomial with $n = 4$ and $p = 0.9$, and the distribution of Y is binomial with $n = 4$ and $p = 0.08$. However, because only four bits are being rated, the possible values of X and Y are restricted to the points shown in the graph in Fig. 5-1. Although the possible values of X are 0, 1, 2, 3, or 4, if $y = 3$, $x = 0$ or 1. By specifying the probability of each of the points in Fig. 5-1, we specify the joint probability distribution of X and Y . Similarly to an individual random variable, we define the range of the random variables (X, Y) to be the set of points (x, y) in two-dimensional space for which the probability that $X = x$ and $Y = y$ is positive.

Figure 5-1 Joint probability distribution of X and Y in Example 5-1



5-1.2 Marginal Probability Distributions

If more than one random variable is defined in a random experiment, it is important to distinguish between the joint probability distribution of X and Y and the probability distribution of each variable individually. The individual probability distribution of a random variable is referred to as its **marginal probability distribution**. In Example 5-1, we mentioned that the marginal probability distribution of X is binomial with $n = 4$ and $p = 0.9$ and the marginal probability distribution of Y is binomial with $n = 4$ and $p = 0.08$.

In general, the marginal probability distribution of X can be determined from the joint probability distribution of X and other random variables. For example, to determine $P(X = x)$, we sum $P(X = x, Y = y)$ over all points in the range of (X, Y) for which $X = x$. Subscripts on the probability mass functions distinguish between the random variables.

EXAMPLE 5-3

The joint probability distribution of X and Y in Fig. 5-1 can be used to find the marginal probability distribution of X . For example,

$$\begin{aligned} P(X = 3) &= P(X = 3, Y = 0) + P(X = 3, Y = 1) \\ &= 0.0583 + 0.2333 = 0.292 \end{aligned}$$

Definition

If X and Y are discrete random variables with joint probability mass function $f_{XY}(x, y)$, then the **marginal probability mass functions** of X and Y are

$$f_X(x) = P(X = x) = \sum_{R_y} f_{XY}(x, y) \quad \text{and} \quad f_Y(y) = P(Y = y) = \sum_{R_x} f_{XY}(x, y) \quad (5-2)$$

where R_x denotes the set of all points in the range of (X, Y) for which $X = x$ and R_y denotes the set of all points in the range of (X, Y) for which $Y = y$



Mean and Variance from Joint Distribution

If the marginal probability distribution of X has the probability mass function $f_X(x)$, then

$$\begin{aligned} E(X) = \mu_X &= \sum_x x f_X(x) = \sum_x x \left(\sum_{R_x} f_{XY}(x, y) \right) = \sum_x \sum_{R_x} x f_{XY}(x, y) \\ &= \sum_R x f_{XY}(x, y) \end{aligned} \quad (5-3)$$

and

$$\begin{aligned} V(X) = \sigma_X^2 &= \sum_x (x - \mu_X)^2 f_X(x) = \sum_x (x - \mu_X)^2 \sum_{R_x} f_{XY}(x, y) \\ &= \sum_x \sum_{R_x} (x - \mu_X)^2 f_{XY}(x, y) = \sum_R (x - \mu_X)^2 f_{XY}(x, y) \end{aligned}$$

where R_x denotes the set of all points in the range of (X, Y) for which $X = x$ and R denotes the set of all points in the range of (X, Y)



In Example 5-1, $E(X)$ can be found as

$$\begin{aligned} E(X) &= 0[f_{XY}(0, 0) + f_{XY}(0, 1) + f_{XY}(0, 2) + f_{XY}(0, 3) + f_{XY}(0, 4)] \\ &\quad + 1[f_{XY}(1, 0) + f_{XY}(1, 1) + f_{XY}(1, 2) + f_{XY}(1, 3)] \\ &\quad + 2[f_{XY}(2, 0) + f_{XY}(2, 1) + f_{XY}(2, 2)] \\ &\quad + 3[f_{XY}(3, 0) + f_{XY}(3, 1)] \\ &\quad + 4[f_{XY}(4, 0)] \\ &= 0[0.0001] + 1[0.0036] + 2[0.0486] + 3[0.02916] + 4[0.6561] = 3.6 \end{aligned}$$

Alternatively, because the marginal probability distribution of X is binomial,

$$E(X) = np = 4(0.9) = 3.6$$



Example 5.2 The joint PMF of two random variables X and Y is given by

$$p_{XY}(x, y) = \begin{cases} k(2x + y) & x = 1, 2; y = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

- What is the value of k ?
- Find the marginal PMFs of X and Y .
- Are X and Y independent?

Solution (a) To evaluate k , we remember that

$$\sum_x \sum_y p(x, y) = \sum_{x=1}^2 \sum_{y=1}^2 k(2x + y) = 1.$$

Thus,

$$\begin{aligned} \sum_{x=1}^2 \sum_{y=1}^2 k(2x + y) &= k \sum_{x=1}^2 \{(2x + 1) + (2x + 2)\} \\ &= k\{(2 + 1) + (2 + 2) + (4 + 1) + (4 + 2)\} \\ &= 18k = 1 \end{aligned}$$

This gives $k = 1/18$.



(b) The marginal PMFs are

$$\begin{aligned} p_X(x) &= \sum_y p_{XY}(x, y) = \frac{1}{18} \sum_{y=1}^2 (2x + y) = \frac{1}{18} \{(2x + 1) + (2x + 2)\} \\ &= \frac{1}{18} (4x + 3) \quad x = 1, 2 \end{aligned}$$

$$\begin{aligned} p_Y(y) &= \sum_x p_{XY}(x, y) = \frac{1}{18} \sum_{x=1}^2 (2x + y) = \frac{1}{18} \{(2 + y) + (4 + y)\} \\ &= \frac{1}{18} (2y + 6) \quad y = 1, 2 \end{aligned}$$

(c) Since $p_X(x)p_Y(y) \neq p_{XY}(x, y)$, we conclude that X and Y are not independent.



Marginal probability

joint probability : $P(X = 0, Y = 1) = 0.1$

		disease X	
		0	1
symptoms Y	0	0.5	0.1
	1	0.1	0.3

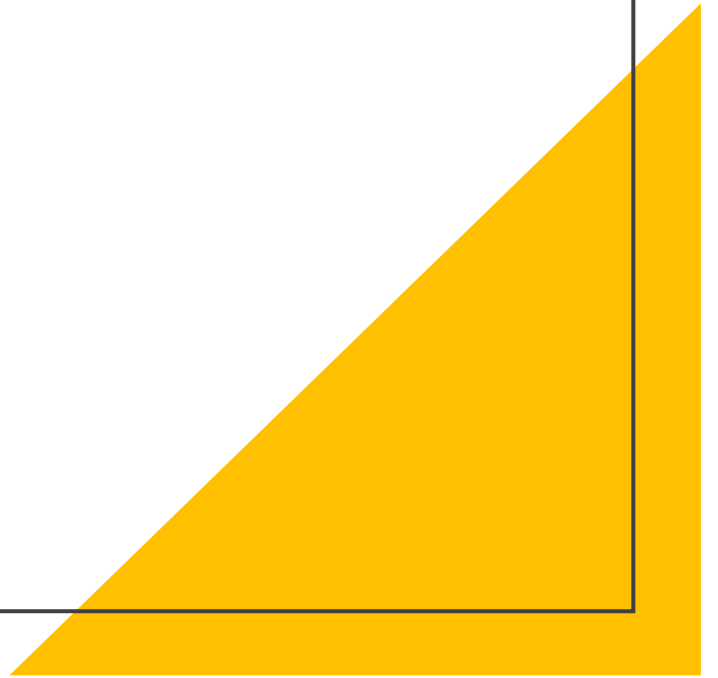
$$\sum_{x,y} P(X = x, Y = y) = 1$$

$$P(Y = 1) = 0.1 + 0.3 = 0.4$$

$$P(X = 0) = 0.1 + 0.5 = 0.6$$

$$P(X = x) = \sum_y P(X = x, Y = y)$$

4. Covariance and Correlation





5-5 COVARIANCE AND CORRELATION

When two or more random variables are defined on a probability space, it is useful to describe how they vary together; that is, it is useful to measure the relationship between the variables. A common measure of the relationship between two random variables is the **covariance**. To define the covariance, we need to describe the expected value of a function of two random variables $h(X, Y)$. The definition simply extends that used for a function of a single random variable.

Definition

The covariance between the random variables X and Y , denoted as $\text{cov}(X, Y)$ or σ_{XY} , is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y \quad (5-28)$$

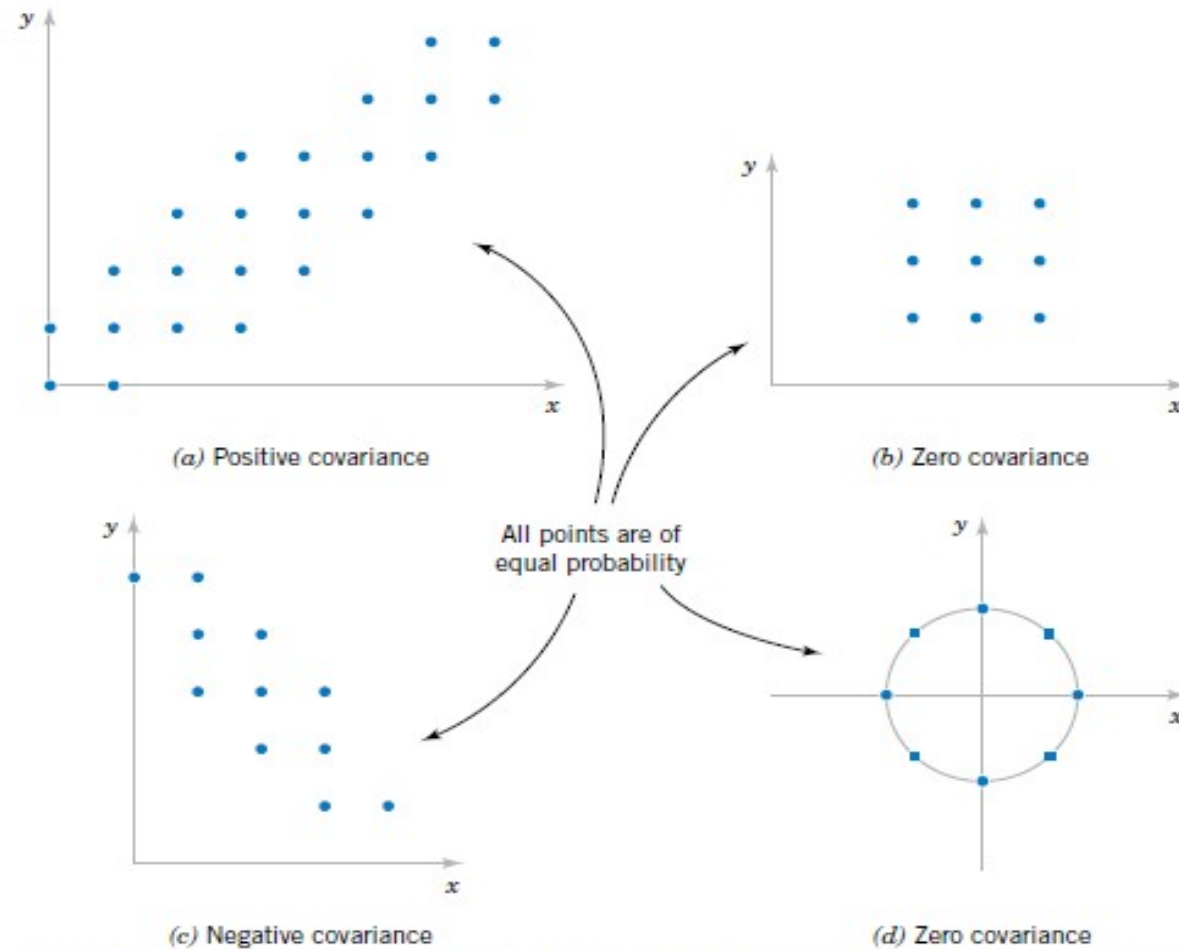


Figure 5-13 Joint probability distributions and the sign of covariance between X and Y .



Let x be a uniformly distributed random variable in the interval $[-1, 1]$. Let y be another random variable given by

. Find the correlation between two variables. Expectation of the random variable x is 0 (since it is uniform in interval $[-1, 1]$). Covariance between the two random variables is ,

$$E(X - \mu_X)(Y - \mu_Y) = E(XY) - 0 \cdot E(Y) = E(XY)$$

$$E(XY) = E(XY^2) = E(X^3) = \int_{-1}^1 x^3 f_X(x) dx = \frac{1}{2} \int_{-1}^1 x^3 dx = 0$$

This indicates that there is no correlation between variables X & Y , though $Y = X^2$

Now, instead let $Y = aX$, where 'a' is a constant.



$$E(XY) = aE(X^2) = \int_{-1}^1 x^2 f_X(x) dx = \frac{1}{2} \int_{-1}^1 x^2 dx = \frac{1}{3}$$

Since $E(aX) = a E(X)$ (a is constant). Covariance is non zero in this case where two random variables are linearly related. Thus, covariance and correlation is a measure of linear dependence between two variables and does not reveal nonlinear relationship between them.

Covariance is a measure of **linear relationship** between the random variables. If the relationship between the random variables is nonlinear, the covariance might not be sensitive to the relationship. This is illustrated in Fig. 5-13(d). The only points with nonzero probability are the points on the circle. There is an identifiable relationship between the variables. Still, the covariance is zero.

The equality of the two expressions for covariance in Equation 5-28 is shown for continuous random variables as follows. By writing the expectations as integrals,

$$\begin{aligned} E[(Y - \mu_Y)(X - \mu_X)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{XY}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [xy - \mu_X y - x\mu_Y + \mu_X \mu_Y] f_{XY}(x, y) dx dy \end{aligned}$$

Now

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mu_{XY} f_{XY}(x, y) dx dy = \mu_X \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{XY}(x, y) dx dy \right] = \mu_X \mu_Y$$

Therefore,

$$\begin{aligned} E[(X - \mu_X)(Y - \mu_Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy - \mu_X \mu_Y = E(XY) - \mu_X \mu_Y \end{aligned}$$

Definition

The **correlation** between random variables X and Y , denoted as ρ_{XY} , is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X) V(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \quad (5-29)$$

For any two random variables X and Y

$$-1 \leq \rho_{XY} \leq +1 \quad (5-30)$$

The correlation just scales the covariance by the standard deviation of each variable. Consequently, the correlation is a dimensionless quantity that can be used to compare the linear relationships between pairs of variables in different units.



Recall: Covariance

$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{n - 1}$$



Interpreting Covariance

$\text{cov}(X,Y) > 0 \rightarrow$ X and Y are positively correlated

$\text{cov}(X,Y) < 0 \rightarrow$ X and Y are inversely correlated

$\text{cov}(X,Y) = 0 \rightarrow$ X and Y are independent

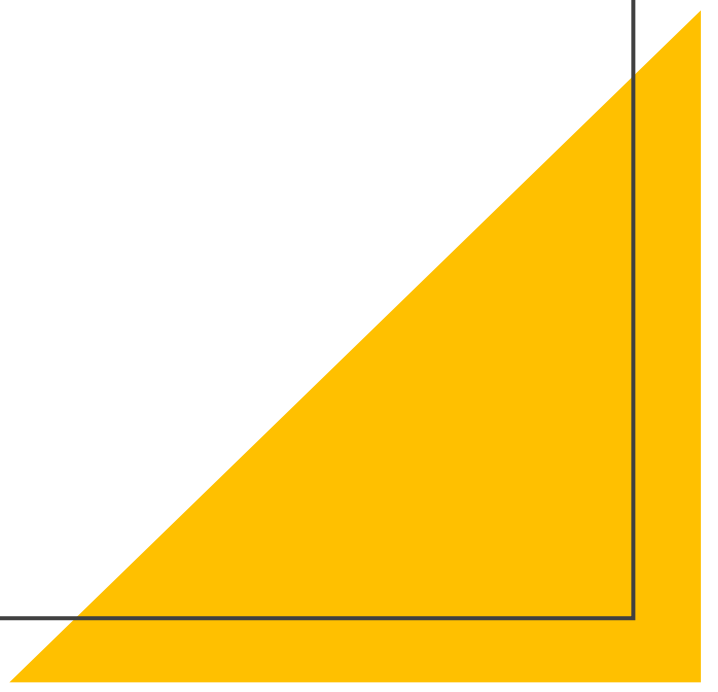


Correlation coefficient

- Pearson's Correlation Coefficient is standardized covariance (unitless):

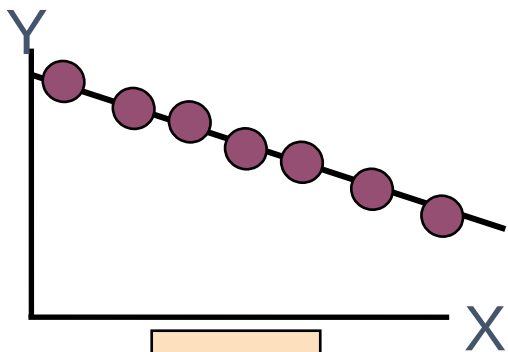
$$r = \frac{\text{covariance}(x, y)}{\sqrt{\text{var } x} \sqrt{\text{var } y}}$$

Correlation

- Measures the relative strength of the *linear* relationship between two variables
 - Unit-less
 - Ranges between -1 and 1
 - The closer to -1 , the stronger the negative linear relationship
 - The closer to 1 , the stronger the positive linear relationship
 - The closer to 0 , the weaker any positive linear relationship
- 
- A large yellow right-angled triangle is positioned in the bottom right corner of the slide, extending from the bottom edge towards the right edge.

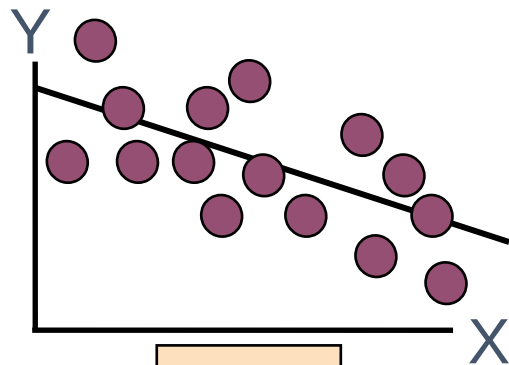


Scatter Plots of Data with Various Correlation Coefficients



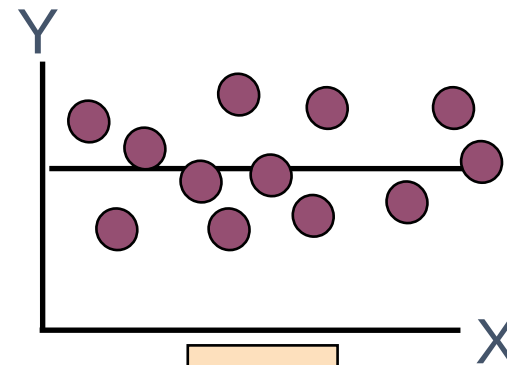
$r =$

-1



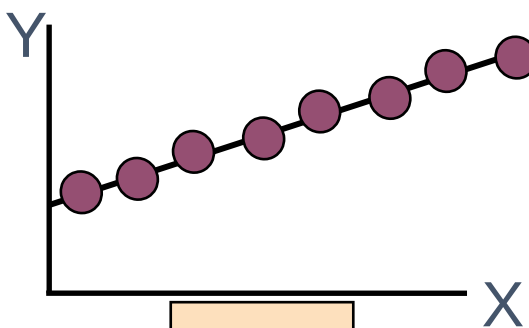
$r =$

-.6



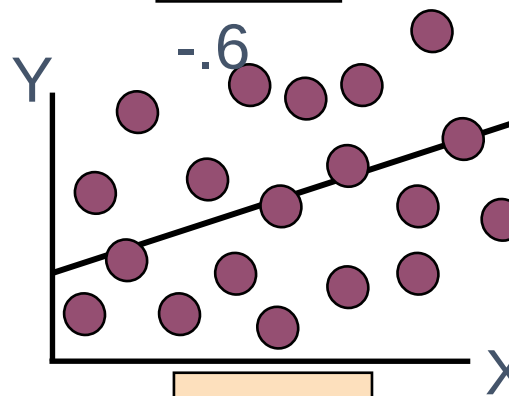
$r =$

0



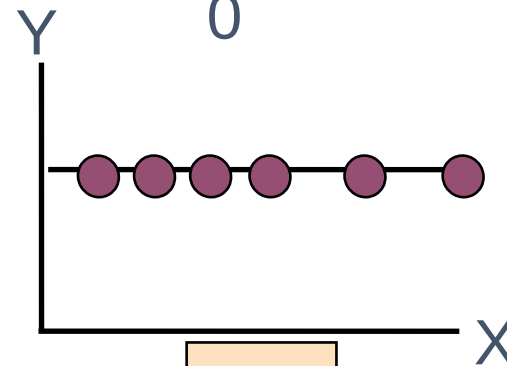
$r =$

+1



$r =$

+.3



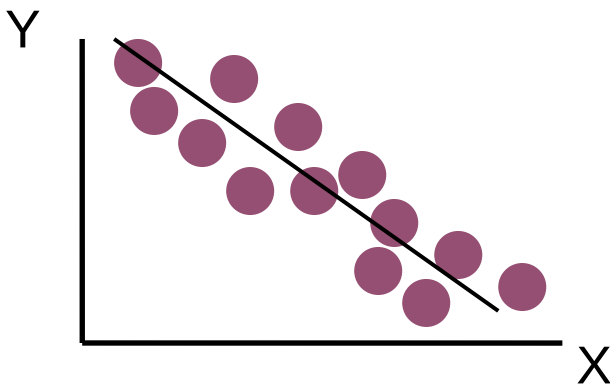
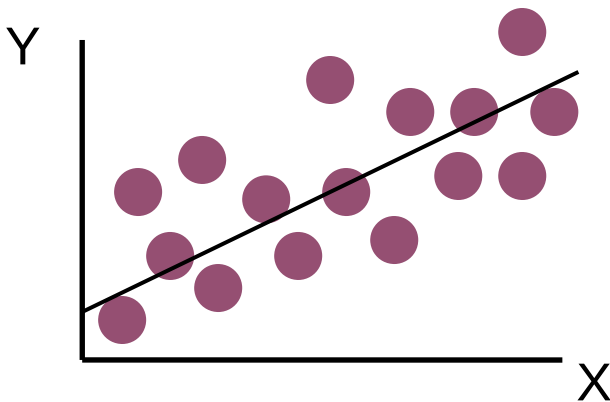
$r =$

0

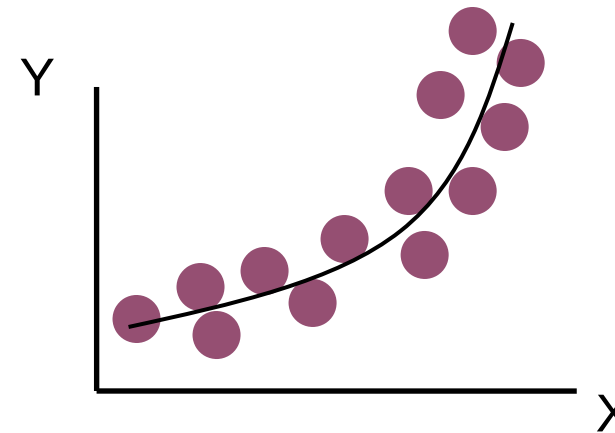
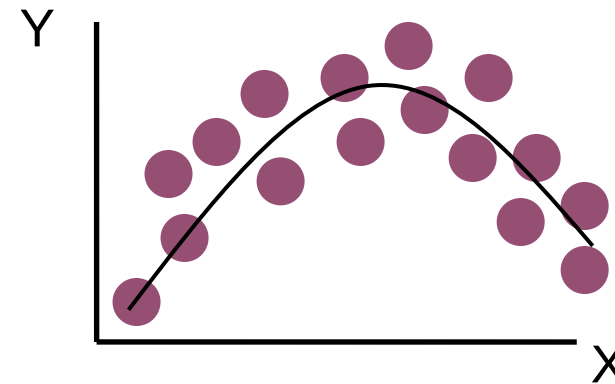


Linear Correlation

Linear relationships

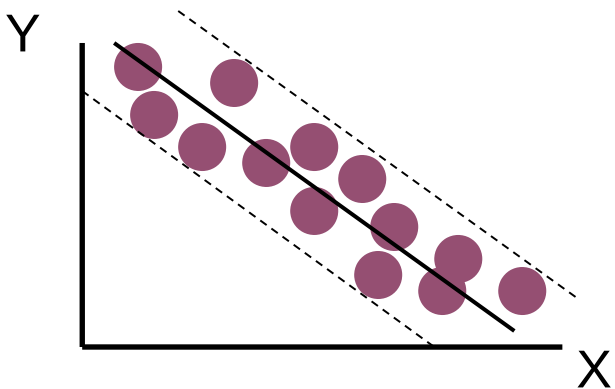
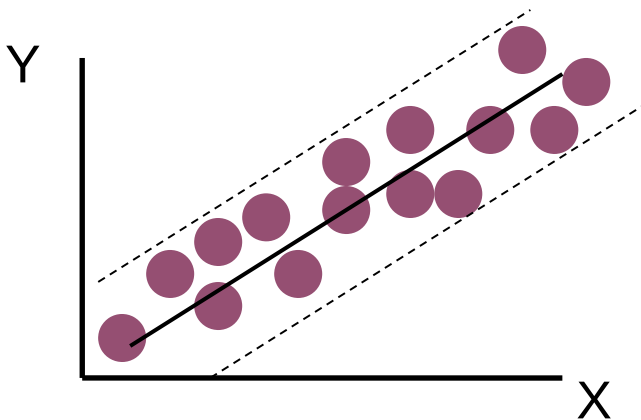


Curvilinear relationships

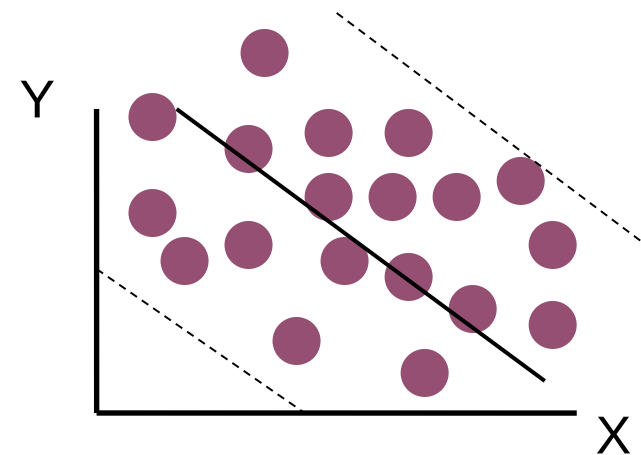
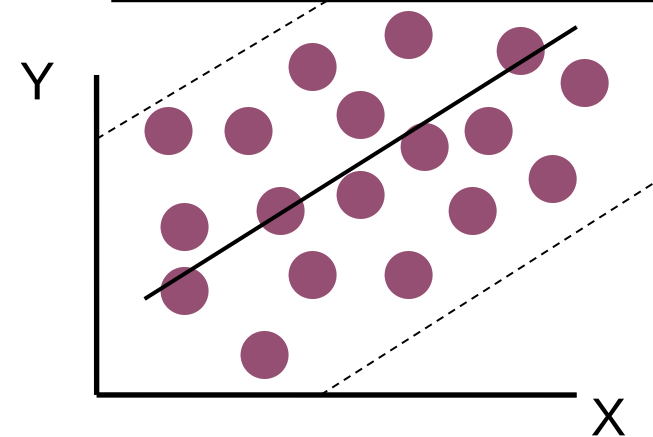


Linear Correlation

Strong relationships



Weak relationships





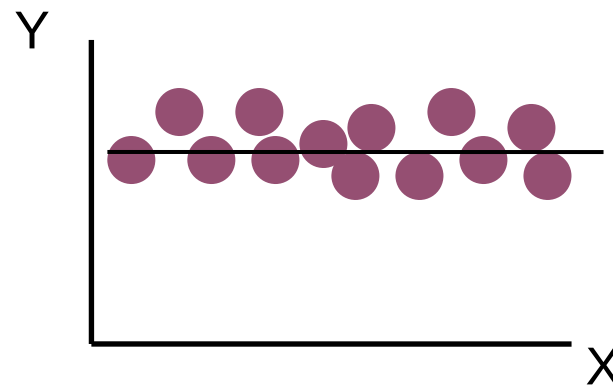
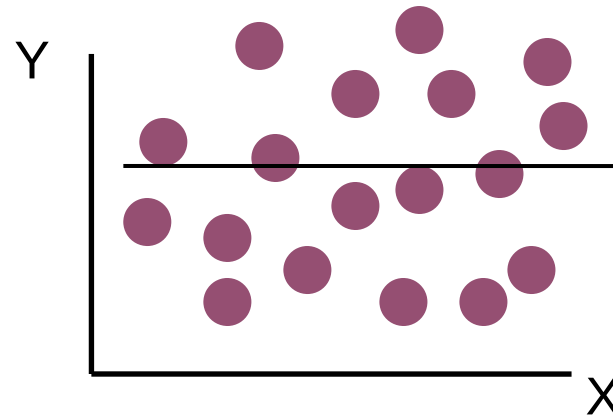
Diamond
2018



Jubilee
2018

Linear Correlation

No relationship



Calculating by hand...

$$\hat{r} = \frac{\text{covariance}(x, y)}{\sqrt{\text{var } x} \sqrt{\text{var } y}} = \frac{\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}}{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}}$$

Simpler calculation formula...

$$\hat{r} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Numerator of
covariance

$$\hat{r} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Numerators of
variance



Properties of Normally distributed variables

- If $x_1, x_2, \dots, x_n$ are independent normal variables, then $y = a_1x_1 + a_2x_2 + \dots + a_nx_n$ is also a normally distributed variable with mean and variance

$$\mu_y = a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n$$

$$\sigma_y^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2$$

- For normally distributed RVs, $\text{cov}(x_1, x_2) = 0 \implies x_1$ and x_2 are independent. In other words, **uncorrelated $\equiv$ independent**.



THANK YOU